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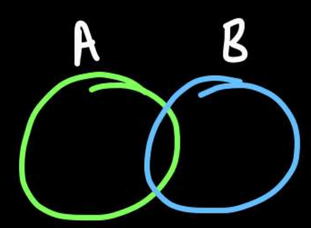
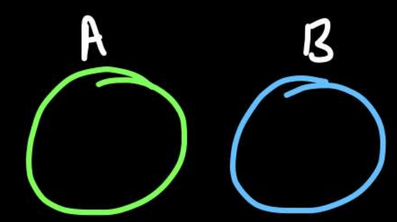
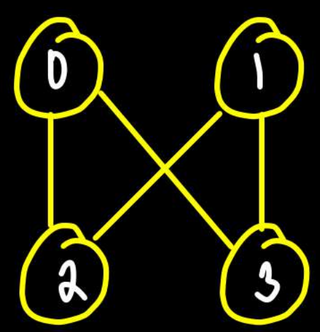
DSU

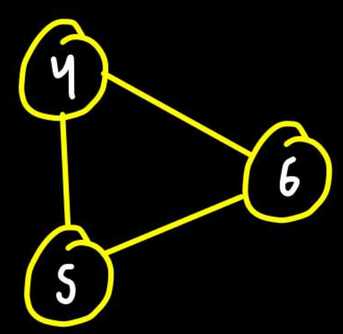
(disjoint set union)

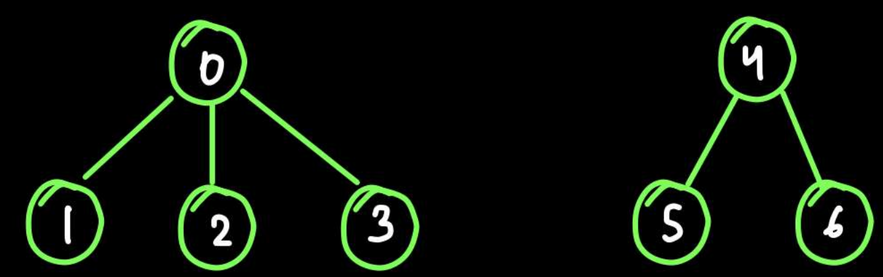




disjoint sets







DSU

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DSU methods

1) make (creating node)

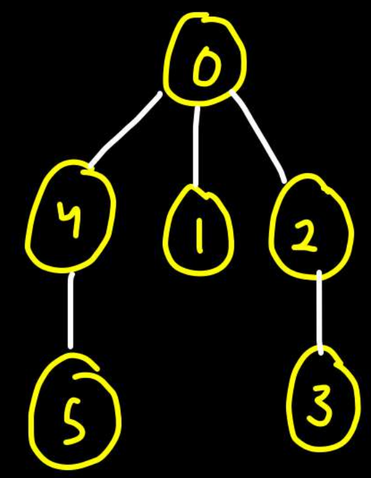
2) find Group leader (root)

3) Union (u,v)

v's leader is attached to u's leader

- ✓ make(0)
- make(1)
- make(2)
- make(3)
- make(4)
- make(5)

✓ union(0,1)
✓ union(2,3)
✓ union(0,2)
✓ findGL(3)
✓ union(4,5)
✓ union(2,5)
0 4



	0	1	2	3	4	5
parent	0	0	0	2	0	4

	0	1	2	3	4	5
size	6	1	2	1	2	1

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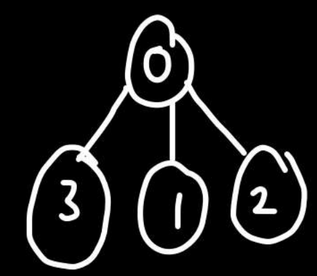
Union by Size

union(1, 0)

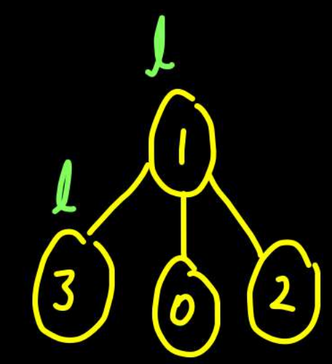
union(2, 1)

union(3, 0)

Worst Case Scenario

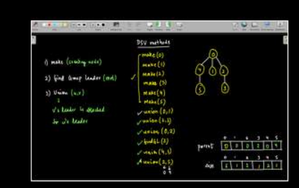


↓ wanted this



size

0	1	2	3
1	4	1	1



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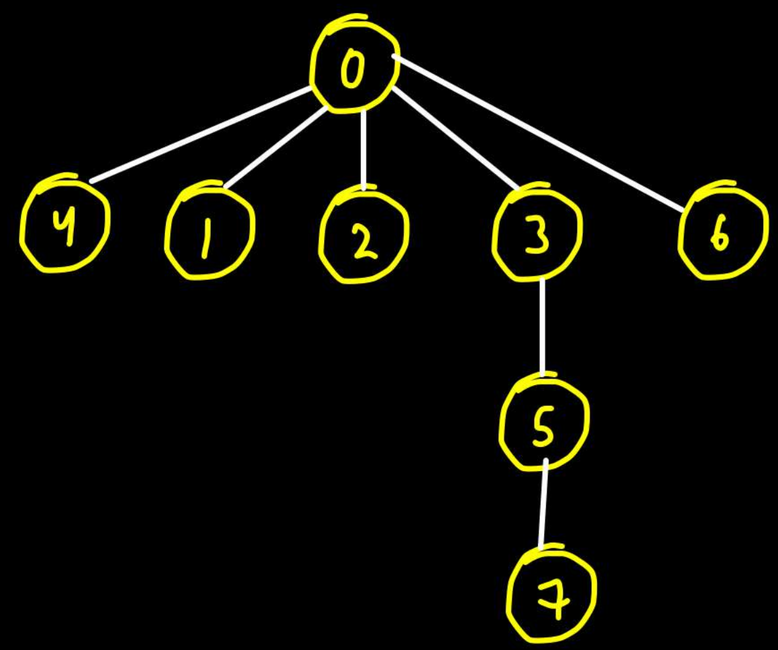
Path Compression



leader(6)

```
int groupLeader(int n){
    if(parent[n]==n) return n;
    int leader = groupLeader(parent[n]);
    parent[n] = leader;
    return leader;
}
```

```
int groupLeader(int n){
    if(parent[n]==n) return n;
    int leader = groupLeader(parent[n]);
    parent[n] = leader;
    return leader;
}
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int groupLeader(int n){
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```
int groupLeader(int n){
    if(parent[n]==n) return n;
    int leader = groupLeader(parent[n]);
    parent[n] = leader;
    return leader;
}
```

	0	1	2	3	4	5	6	7
parent	0	0	0	0	0	3	4	5

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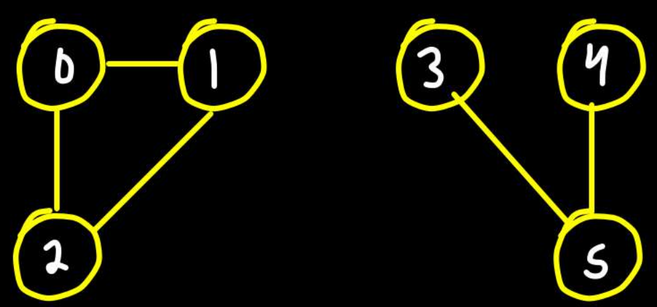
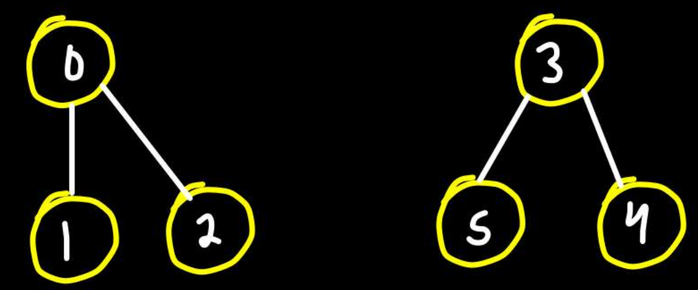
Q. Number of Provinces (DSU) = no. of group leaders

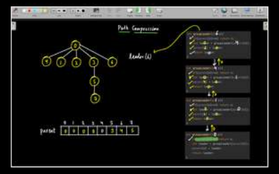




	0	1	2	3	4	5
0	x	1	1	0	0	0
1	1	x	1	0	0	0
2	1	1	x	0	0	0
3	0	0	0	x	0	1
4	0	0	0	0	x	1
5	0	0	0	1	1	x

adj

parent =

0	1	2	3	4	5
0	0	0	3	3	3

ans = 2